

ANGULAR HANDS-ON

❖ HANDS-ON 1

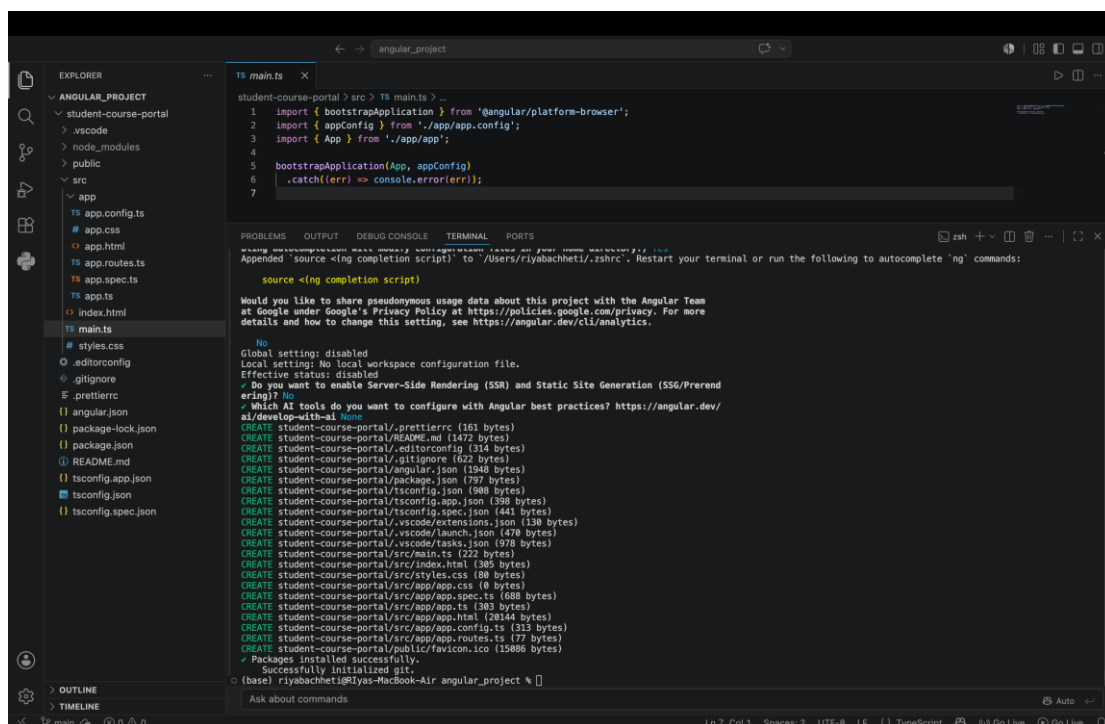
❖ Environment Setup, Project Structure & First Component

Scaffold and Explore the Angular Project

Goal: Create the Angular workspace and understand what each generated file does.

Steps:

1. Run `ng new student-course-portal --routing --style=css`. Accept defaults for all prompts.



2. Open the project in VS Code. For each of the following files, write a one-line comment in a notes.txt file explaining its purpose: angular.json, tsconfig.json, tsconfig.app.json, package.json, src/main.ts, src/app/app.module.ts (or app.config.ts in standalone mode), src/app/app.component.ts, src/index.html.

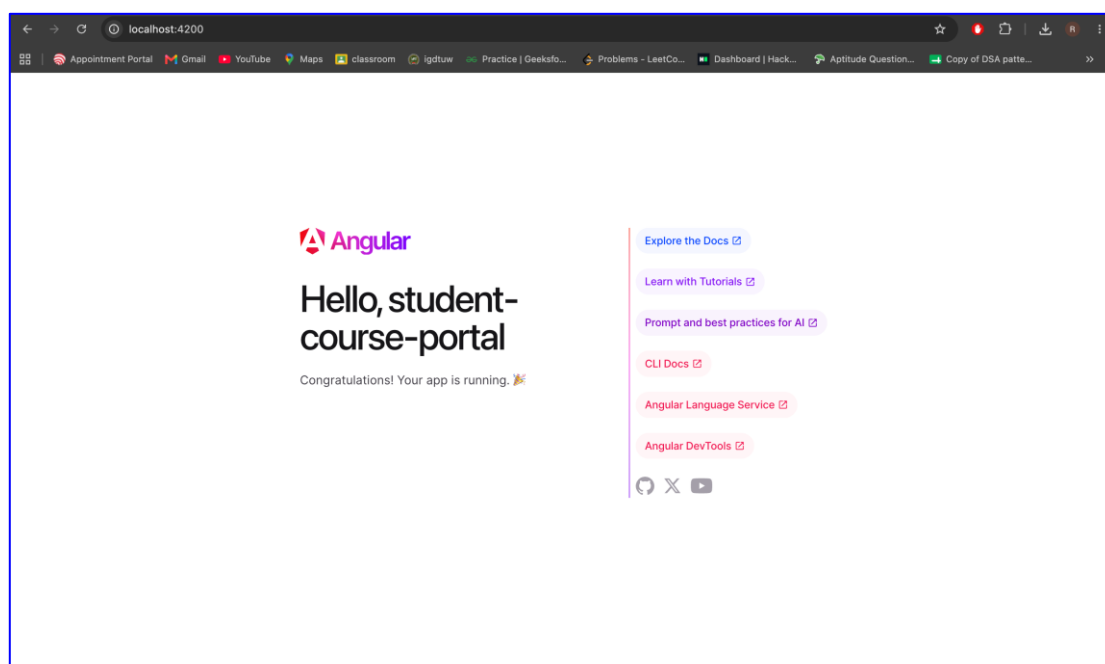
```
notes.txt U X
student-course-portal > notes.txt
1  angular.json
2  Configures how Angular builds, serves, tests, and manages the project.
3
4  tsconfig.json
5  Contains the main TypeScript compiler configuration for the project.
6
7  tsconfig.app.json
8  Contains TypeScript compiler settings specific to the Angular application source code.
9
10 package.json
11 Stores project metadata, dependencies, and npm scripts used to run the application.
12
13 src/main.ts
14 The entry point of the Angular application that bootstraps the root component.
15
16 src/app/app.config.ts
17 Configures application-wide providers and settings for the standalone Angular application.
18
19 src/app/app.ts
20 Defines the root component that controls the main user interface of the application.
21
22 src/index.html
23 The main HTML page that hosts the Angular application.
24
```

3. Run `ng serve` and open `http://localhost:4200`. Verify the default Angular welcome page loads.

```
Successfully initialized git.
● (base) riyabachheti@Riyas-MacBook-Air angular_project % cd student-course-portal
○ (base) riyabachheti@Riyas-MacBook-Air student-course-portal % ng serve
Initial chunk files | Names | Raw size
main.js | main | 47.84 kB
styles.css | styles | 95 bytes
| Initial total | 47.94 kB

Application bundle generation complete. [0.939 seconds] - 2026-07-24T08:59:01.594Z

Watch mode enabled. Watching for file changes...
NOTE: Raw file sizes do not reflect development server per-request transformations.
→ Local: http://localhost:4200/
→ press h + enter to show help
```

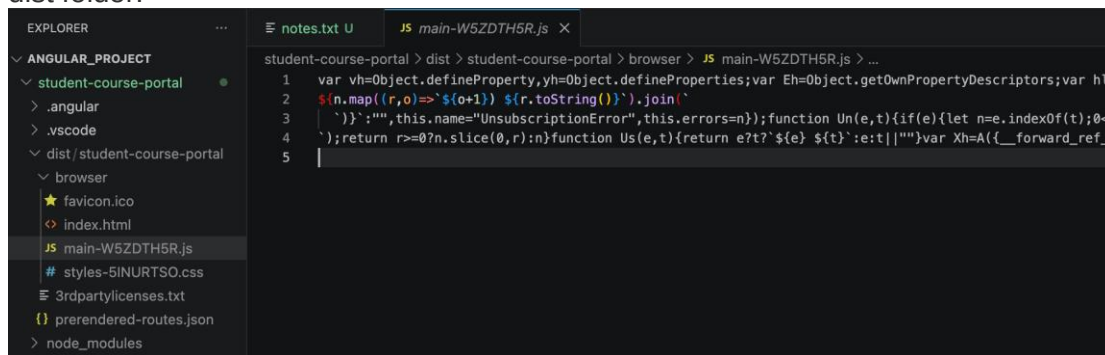


4. Run `ng build` and inspect the `dist/` folder. Note the generated files — identify which file contains the compiled application code.

```
(base) riyabachheti@RIYas-MacBook-Air student-course-portal % ng build
Initial chunk files | Names | Raw size | Estimated transfer size
main-W5ZDTH5R.js | main | 214.73 kB | 58.91 kB
styles-5INURTS0.css | styles | 0 bytes | 0 bytes
Initial total | 214.73 kB | 58.91 kB

Application bundle generation complete. [1.376 seconds] - 2026-07-24T09:03:39.105Z
Output location: /Users/riyabachheti/Documents/Coding/angular_project/student-course-portal/dist/student-course-portal
```

The compiled application code is contained in the generated `main-*.js` file inside the `dist` folder.



```
student-course-portal > dist > student-course-portal > browser > JS main-W5ZDTH5R.js > ...
1 var vh=Object.defineProperty,yh=Object.defineProperties;var Eh=Object.getOwnPropertyDescriptors;var hl=
2 $(n.map((r,o)=>`${o+1})` ${r.toString()}').join('
3 `)`:'"',this.name="UnsubscriptionError",this.errors=n));function Un(e,t){if(e){let n=e.indexOf(t);0<=
4 `);return r==0?n.slice(0,r):n}function Us(e,t){return e?t?`${e}`:${t}`:e:t||''}var Xh=A({__forward_ref_
5 |
```

5. In `angular.json`, locate the budgets configuration under `architect > build > configurations > production`.

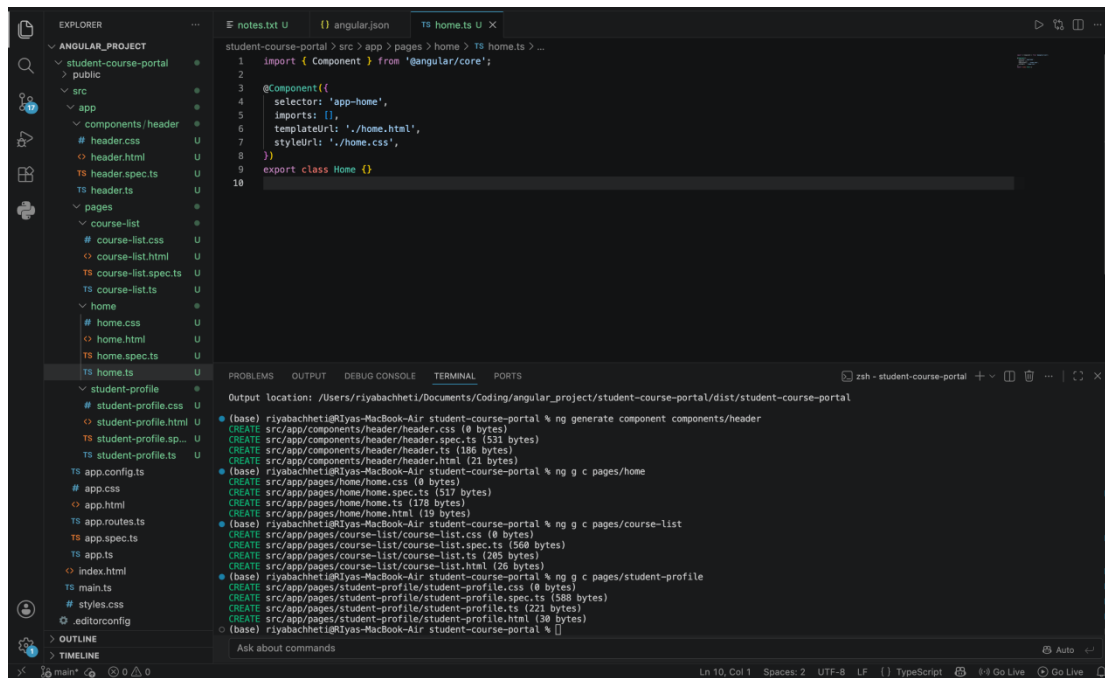
```
notes.txt U {} angular.json X
-portal > {} angular.json > {} projects > {} student-course-portal > {} architect > {} build >
8   "projects": {
9     "student-course-portal": {
15      "architect": {
16        "build": {
31          "configurations": {
32            "production": {
33              "budgets": [
34                {
35                  "type": "initial",
36                  "maximumWarning": "500kB",
37                  "maximumError": "1MB"
38                },
39                {
40                  "type": "anyComponentStyle",
41                  "maximumWarning": "4kB",
42                  "maximumError": "8kB"
43                }
44              ],
45              "outputHashing": "all"
46            },
47            "development": {
48              "optimization": false,
49              "extractLicenses": false,
50              "sourceMap": true
          }
        }
      }
    }
  }
```

Task 2: Create and Organise Components

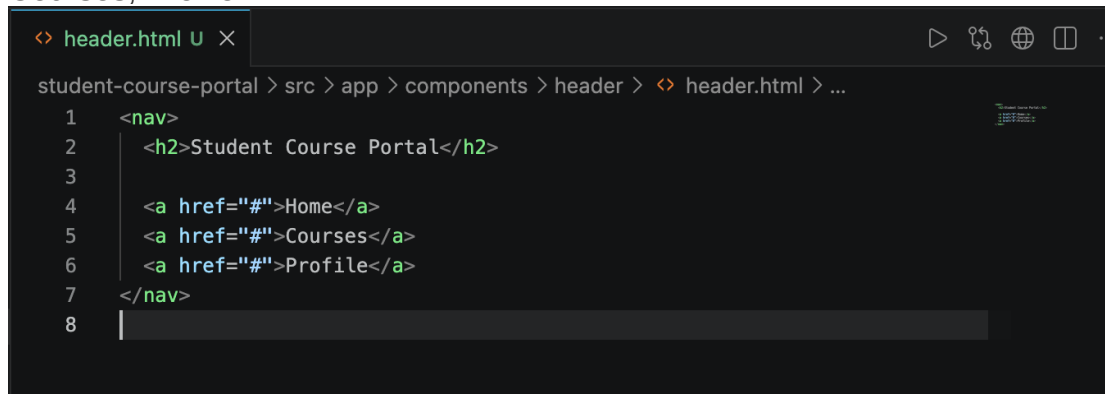
Goal: Generate the core page components for the Student Course Portal.

Steps:

6. Generate four components using Angular CLI: ng generate component components/header, ng generate component pages/home, ng generate component pages/course-list, ng generate component pages/student-profile. Observe the 4 files generated for each: .ts, .html, .css, .spec.ts.



7. In HeaderComponent's template (header.component.html), add a `<nav>` with the portal name 'Student Course Portal' and placeholder links: Home, Courses, Profile.



8. In HomeComponent's template, add a welcome heading (`<h1>`), a short description paragraph, and a stats row showing three hardcoded values: Courses Available: 12, Enrolled: 3, GPA: 3.8.

```
<> home.html U X
student-course-portal > src > app > pages > home > <> home.html > ...
1  <h1>Welcome to Student Course Portal</h1>
2
3  <p>
4    Manage your courses and student information.
5  </p>
6
7  <div>
8    <p><strong>Courses Available:</strong> 12</p>
9    <p><strong>Enrolled:</strong> 3</p>
10   <p><strong>GPA:</strong> 3.8</p>
11 </div>
12
13
```

9. Add `<app-header>` and `<router-outlet>` to `app.component.html`, replacing the default boilerplate content.

```
<> app.html M X
student-course-portal > src > app > <> app.html > ...
> Find Aa ab .* No results ↑ ↓ ≡ ×
1  <app-header></app-header>
2
3  <router-outlet></router-outlet>
4
5
```

10. Verify ng serve still compiles without errors and the header and home page content are visible.

